

Optimization of Wagon Maintenance Mix Allocation in Freight Railways Using Integer Linear Programming

Otimização do Mix de Alocação da Manutenção de Vagões em Ferrovias de Carga por Programação Linear Inteira

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Versão PT

ABSTRACT

We propose a **multi-period integer linear programming** model that allocates regional wagon-maintenance demand among heterogeneous posts, considering wagon type, technical eligibility, effective person-hour (HH) capacity, composite costs, and backlog. On an anonymized, reproducible instance (1,004 interventions, 12 months), the optimal plan cut final backlog from 39 to 9 wagons and total system cost by 25.4% versus a local-first heuristic, solving in under 1 s.

Keywords: freight railways; wagon maintenance; integer linear programming.

WHY IT MATTERS

Brazilian railways moved more than **540 million net tons** in 2024. In fleets of about **20,000 wagons** with ~18,000 interventions a year, choosing **where** and **when** to service each wagon drives cost, fleet availability and post occupancy. Aggregate capacity slack does not guarantee service: capacity is scope-, post- and month-specific.

Method: Python + SciPy · HiGHS MILP solver (gap 10^{-4}) · open code and data via the QR code at the bottom.

1. Problem

- Light scopes (**VR1**, **VR2**) and corrective work (**VMC**) can go to any post;
- Heavy reviews (**VR4**, **VRG**) need equipment available only at **PM2**;
- Sending a wagon to a distant post adds logistics and unavailability cost — sometimes **postponing is better**;
- Monthly capacity drops with staff leave and equipment repair.

Research questions (α β γ mark where each one is answered)

- α** How can maintenance demand be allocated among posts at minimum total operating cost without violating operating constraints?
- β** How much does jointly considering capacity, technical eligibility and composite costs improve planning over local rules or heuristics?
- γ** How much does an optimization model raise the analytical quality and managerial gains of tactical maintenance planning?

3. Test instance

- 1,004** interventions · **12** months · **3** regions · **3** posts
- Gondolas **GDT** and **GDU**; difficulty index from gross weight: $\kappa_{GDU} = 150/130 = 1.154$
- 1,152** integer variables · **396** constraints
- Annual workload 14,556 HH vs. 15,664 effective HH (7.6% slack) — yet heavy scopes need **6,593 HH** against **6,525 HH** at PM2: **a local bottleneck**.

Scope	PM1	PM2	PM3	HH GDT	HH GDU
VR1 (light)	✓	✓	✓	8	9
VR2 (light)	✓	✓	✓	12	14
VR4 (heavy)	✗	✓	✗	30	35
VRG (heavy)	✗	✓	✗	40	46
VMC (corrective)	✓	✓	✓	10	12

Technical eligibility and person-hours per wagon.

4. Baseline and ablation β

Local-first heuristic under the **same** demand, eligibility, minimum/maximum capacity and backlog rules:

① new demand + backlog → ② minimum workload of each post → ③ **local** eligible post → ④ other posts by increasing cost → ⑤ remainder goes to backlog.

Same inputs and constraints for both methods: they differ in decision quality, not in the rules they obey.

Scenario	Cost (MU)	Final backlog	Remote
Local-first heuristic	228,672	39	306
Full ILP	170,690	9	401
Direct cost only	178,447	9	612
No minimum workload	170,591	9	398
LP relaxation	169,297	8.68	—

Row 1: baseline. Ablation: ignoring logistics and unavailability adds transfers (401→612) and real cost. LP relaxation yields fractional wagons (gap 0.82%).

5. Results — heuristic vs. optimized β γ

39→9

final backlog (wagons)

−54.4%

cumulative backlog

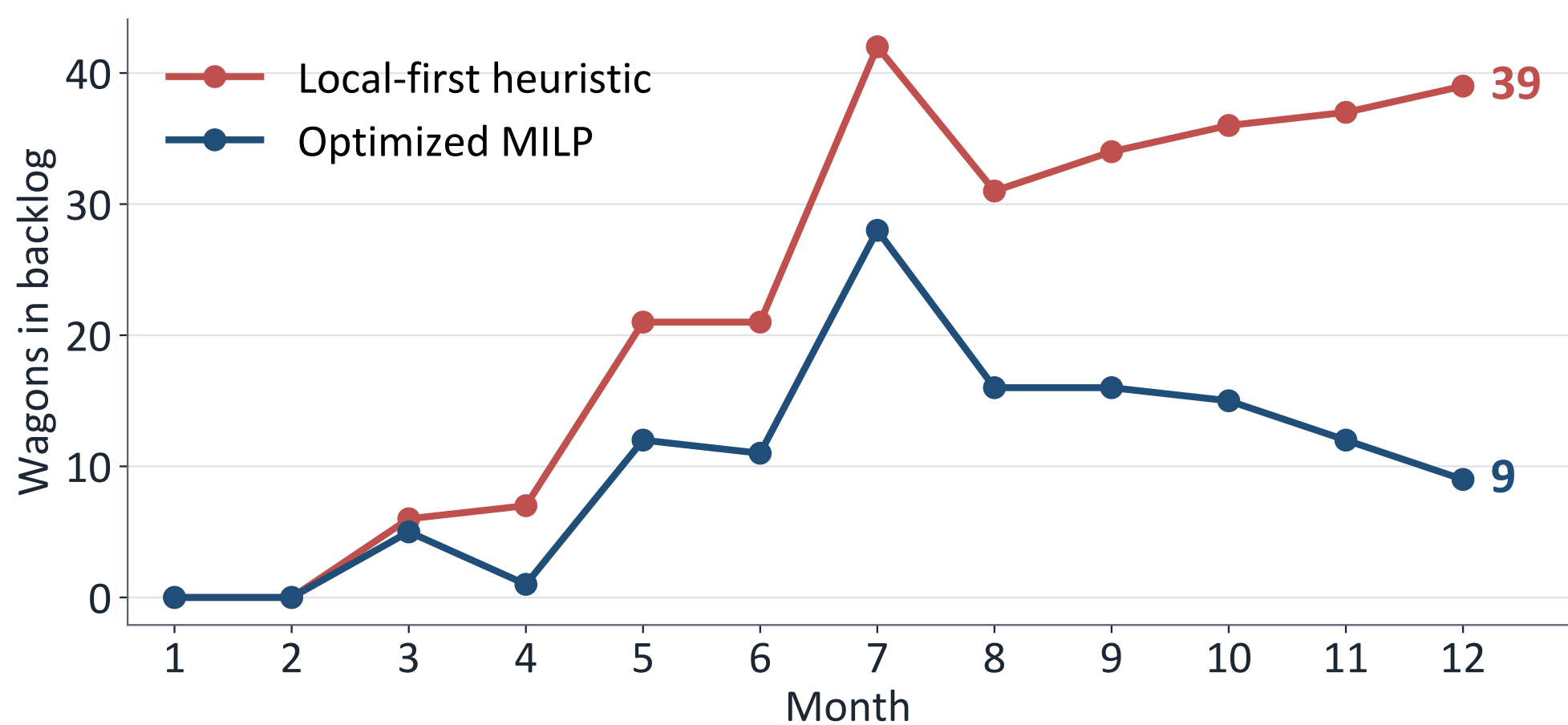
−25.4%

total system cost

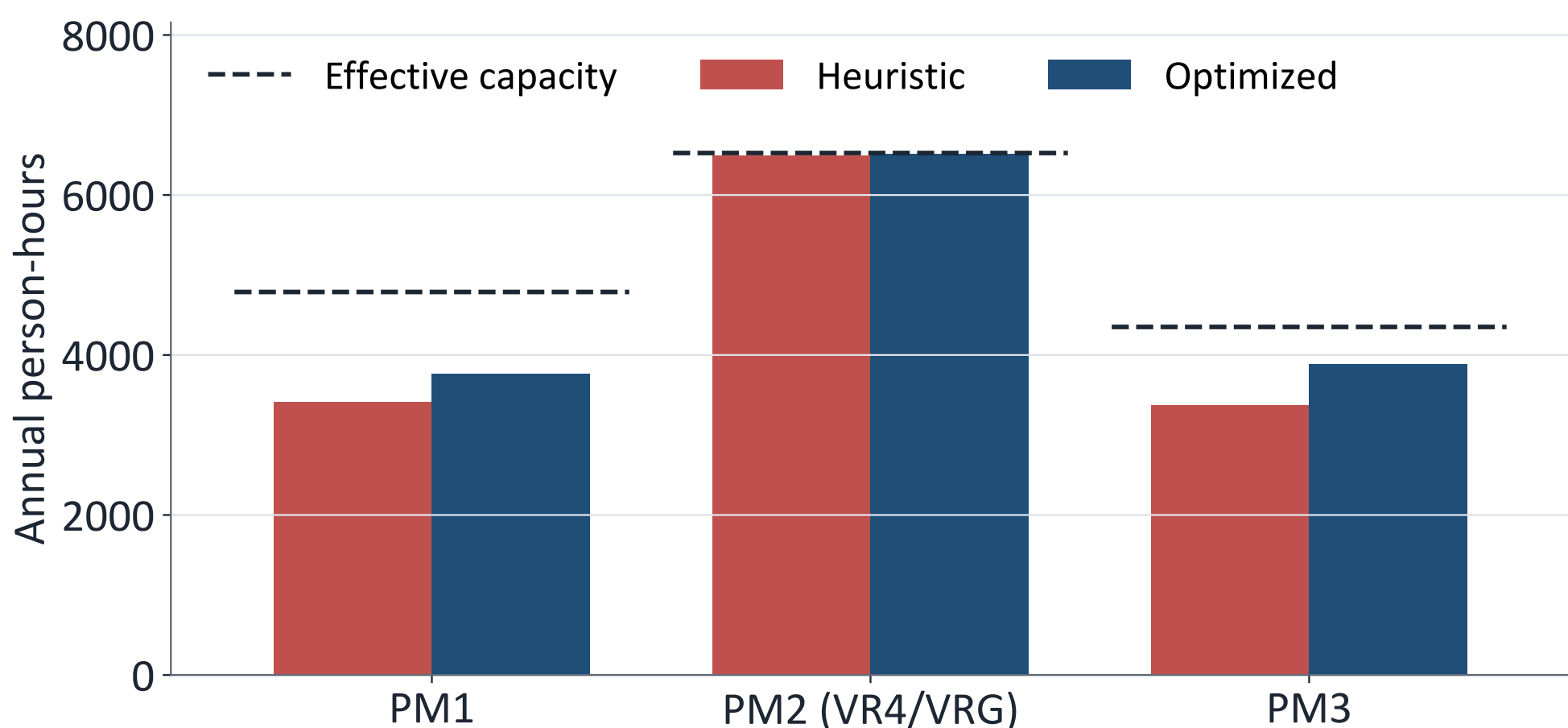
<1 s

solve time (1,152 var.)

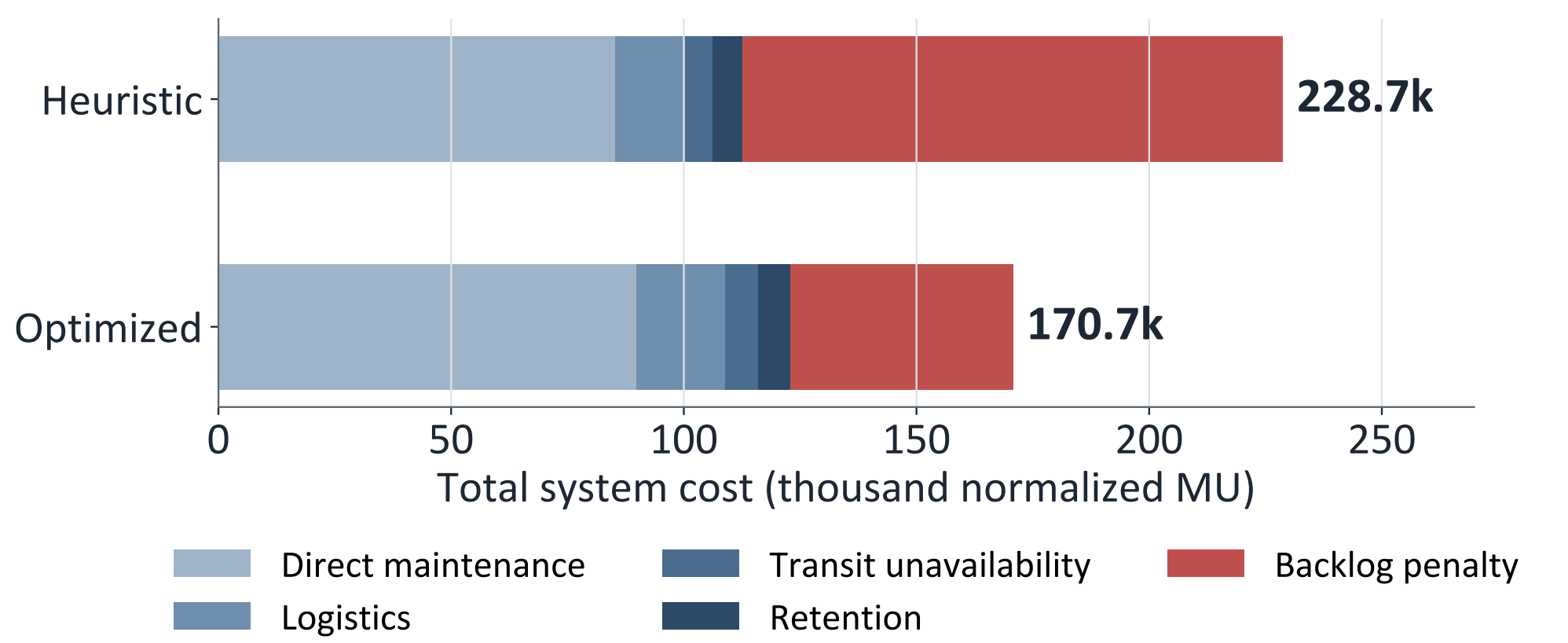
11,520 var. < 6 s



(a) Wagons in backlog per month.



(b) Annual HH per post. The heuristic fills PM2 (the only heavy-scope post) with 119 light wagons; the optimum sends 34 and completes 75 VR4 reviews instead of 40.



(c) Cost components: the optimum spends +9% on service to cut 59% of the backlog penalty.

6. Conclusions

- α** The flow-balance formulation separates **technical infeasibility** from **economically justified postponement**.
- β** **Source of the gain:** freeing the PM2 bottleneck from light work; VR4 backlog drops from 216 to 41 wagon-months. Transfers happen only when cheaper than waiting.
- Aggregate HH slack ≠ serviceable demand: capacity is scope-, month- and region-specific.
- γ** Solves in seconds and supports outage, capacity, outsourcing, equipment and movement scenarios — practical for monthly tactical planning.
- Serviced 995 vs. 965 wagons; cumulative backlog 274 → 125 wagon-months.
- Next steps:** demand uncertainty, network capacity, backlog age, MRO inventory.

Key references

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Normalized costs; anonymized instance.

FAQ · code · data



questions & downloads

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